BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. Time period is reciprocal of:

(a) Frequency
(b) Wave length
(c) Velocity
(d) Distance

Q2. If mass is attached to a spring then its time period is:

(a) Increases
(b) Decreases
(c) Remains constant
(d) None of these

Q3. In double slits experiment for constructive interference the condition is:

(a) $d \sin\theta = m\lambda$
(b) $d \sin\theta = (m+1/2)\lambda$
(c) $2\, d \sin\theta = m\lambda$
(d) $2\, d \sin^2\theta = (m+1/2)\lambda$

Q4. In Michelson's interferometer, the wave length of light is given as:

(a) $\lambda = x/m$
(b) $\lambda = 2m/m$
(c) $\lambda = 2x/m$
(d) mx^2

Q5. Who proposed wave theory of light:

(a) Kepler
(b) Huygens
(c) Galelio
(d) Issac Newton

Q6. The sound waves having frequency less than 20Hz is called:

(a) Infrasonic
(b) ultrasonic
(c) andiable
(d) None of these

Q7. The unit of power of lens is:

(a) Hertz
(b) Bell
(c) Dyne
(d) Dioeters

Q8. The unit of power of lens is:

(a) Hertz
(b) Bell
(c) Dyne
(d) Dipoters

Q9. The power of a lens be one meter the focal length will be:

(a) 2.5 m
(b) 1 diopter
(c) 3 dipoter
(d) 4 dipoter

Q10. Simple microscope consists of:

(a) One convex lens
(b) two convex lens
(c) One concave lens
(d) Two concave lens

Q11. Torque is also called:

(a) Moment of force
(b) Momentum
(c) Moment arm
(d) None of these

Q12. It is the perpendicular distance from the axis of rotation to the direction of force, is called _____.

(a) Moment arm
(b) Moment of force
(c) Angular momentum
(d) None of these

Q13. The magnitude of angular momentum is:

(a) e Fsin θ
(b) Fd sin θ
(c) rP sin θ
(d) Fv sinθ

Q14. The weight of an object would be minimum, when it is placed at the:

(a) Surface of earth
(b) Centre of earth
(c) Equator
(d) Pole

Q15. The approximate value of the average density of earth is:

(a) 5.5 x 103 Kg/m^3
(b) 6.5 x 103 Kg/m^3
(c) 7.5 x 105 Kg/m^3
(d) 8.5 x 103 Kg/m^2

Q16. The value fo gravitational constant G is:

(a) 6.63 x 10^-34 N.m^2/Kg^2
(b) 6.2 x 10^-24 Nm^2/Kg^2
(c) 6.67 x 10-11 N.m2/Kg2
(d) 6.67 x 10^11 Nm/Kg^2

Q17. Pull of the earth on mass of 20 Kg on the surface fo earth is:

(a) 20N
(b) 19.6 N
(c) 196 N
(d) 1.96 N

Q18. Power is defined as, the rate of doing work and is expressed as :

(a) P x W x t
(b) P = ΔW/Δt
(c) P = F.V
(d) Both (a) (c)

Q19. ½ horse power = _____

(a) 746 Watts
(b) 726 Watts
(c) 373 Watts
(d) 363 Watts

Q20. S.I Unit of work is:

(a) It possesses maximum P..E
(b) Newton second
(c) Newton cm
(d) None of these

Q21. A body is performing SHM, at the mean position:

(a) It possesses maximum P.E
(b) Its acceleration is maximum
(c) Its acceleration is zero
(d) Its velocity is zero

Q22. Acceleration due to gravity near the earth surface is _____ if air friction is ignored.

(a) 980 cm/sec^2
(b) 32 ft/sec^2
(c) 9.8 m/sec^2
(d) All of these

Q23. Unit of momentum is:

(a) Joule
(b) Watt
(c) Kg m/sec
(d) Newton

Q24. Self adjusting force is:

(a) Friction
(b) Gravity
(c) Centripetal force
(d) Weight

Q25. In the absence of air resistance the horizontal component of velocity _____ during projectile motion:

(a) Increases
(b) Decreases
(c) Remains constant
(d) None of these

Q26. To get maximum horizontal range, the projectile must be launched at an angle of:

(a) 15o
(b) 45o
(c) 60o
(d) 90o

Q27. One radian is the angle subtended at the centre by an arc exactly time graph

(a) Half of the circumference of circle
(b) Diameter of circle
(c) Circumference of circle
(d) Radius of circle

Q28. The projectile moves with uniform velocity, then the distance time graph is:

(a) Hyperbola
(b) Parabola
(c) curve
(d) straight line

Q29. Which of the following types of force cannot do work on a body?

(a) centripetal force
(b) Elastic force
(c) Gravitational force
(d) Coulomb force

Q30. When body moves along a circular path its velocity:

(a) Remain same
(b) becomes zero
(c) some time changes
(d) changes continuously

Q31. The force needed to move a body around a circular a circular path is called:

(a) Centripetal force
(b) Centripetal acceleration
(c) Centrifugal force
(d) Gravitational force

Q32. The turning effect of force is called:

(a) Angular momentum
(b) Torque
(c) Linear momentum
(d) Work

Q33. The branch of physics which deals with the atomic nuclei is called:

(a) Solid state physics
(b) Medical physics
(c) Nuclear physics
(d) None of these

Q34. Dimensions of work are:

(a) MLT^2
(b) M^2LT^2
(c) ML^2T^2
(d) ML^2T^-2

Q35. The vector product of two equal vector is:

(a) A vector
(b) Null vector
(c) square of vector
(d) Unit vector

Q36. MT^-2L^2 is the dimensions of :

(a) Velocity
(b) Momentum
(c) Power
(d) Torque

Q37. If A = 10 N East, then -A = -10N:

(a) West
(b) North
(c) South
(d) None of these

Q38. The scalar product of two vectors A and B can be written as:

(a) A.B = AB cosθ
(b) A.B = AB cosθ
(c) A.B = AB sinθ
(d) None of these

Q39. Null vector is a vector having zero magnitude and:

(a) Specific direction
(b) Arbitrary direction
(c) No director
(d) Opposite director

Q40. The magnitude of vector 2i + 3j + 6k is:

(a) 11
(b) 36
(c) 7
(d) 9

Q41. If A = Axi + Ayj + Azk and B = Bxi + Byj + Bzk, then:

(a) A x Bx - AyBy - AzBz
(b) Ax Bx + AyBy + AzBz
(c) Ax By + AyBz + AzBx
(d) AxBz + ayBx + AzBy

Q42. In vector product, a vector multiplied with itself will in a _____ magnitude.

(a) Double
(b) Zero
(c) Half
(d) None of these

Q43. The weight of a body falling freely will be _____

(a) mg
(b) Zero
(c) mg -6πrη
(d) mg +

Section-B Short Answer

> Note: Answer any SIX of the following questions.

Q2. What is equilibrium, state any one condition of equilibrium.

Q3. Prove that vector product does not obey the Commutative Law.

Q4. Find the work done in moving an object a vector r= 3i + 2j - 5k if the applied force is F = 2i + j - k.

Q5. Calculate the average density of earth.

Q6. State and prove the Law of Conservation of Energy.

Q7. State the basic of second pendulum.

Q8. Find the length of second pendulum.

Q9. An object is placed at a distance of 30 cm from a convex lens of focal 10cm. Find the position of image.

Q10. How for a part are the diffracting planes in a NaCl crystal for which x-rays of wave length 1.54Å make glancing angle of 15°C - 54°C 1th order?

Section-C (Descriptive Answer)

> Note: Answer any TWO of the following questions.

Q11. Two unequal masses are attached to a string which passes over a frictionless pulley. If the bodies are moving vertically. Find the expression for acceleration and tension in the string.

Q12. What is simple pendulum? Show that the motion of simple pendulum is SHM. Drive the expression for its time period.

Q13. Write a note on any ONE of the following: Compound Microscope / Michelson's Interferometer / Centripetal aacceleration